Dynamical Systems

a structural cheatsheet · after C. Kühn, *Foundations of Applied Dynamical Systems* (TUM)

Stability of equilibria §2

0 Equilibrium (fixed point) $\varphi_t(x^*) = x^* \quad \forall t \in T$
ODE: $f(x^*) = 0$; map: $g(x^*) = x^*$

1 Lyapunov stable $\forall U\ \exists V \subseteq U:\ \varphi_t(V) \subseteq U\ \ \forall t \geq 0$
nearby orbits stay nearby

2 Attracts a positive-measure set $\mu(B(x^*)) > 0$
basin $B(A) := \{x_0 : \omega(x_0) \subset A\}$

3 Attracts a whole neighborhood $\varphi_t(x_0) \xrightarrow{t\to+\infty} x^*\ \ \forall x_0 \in U$

4 Basin is the whole space $U = X$

- **equilibrium**: saddle, unstable node; $x' = x$ at 0
- **stable**: center $\ddot{x} + x = 0$: stable, *not* asympt. stable
- **Milnor attractor**: $x' = -x^2$: basin $[0,\infty)$ yet unstable — row 1 skipped
- **asympt. stable**: stable node / focus
- **globally asympt. stable**: $x' = -x$ on $\mathbb{R}$

Stability certificates §2

Lyapunov function on $U \ni x^*$
- (D1) L continuous
- (D2) $L(x^*) = 0,\ L(x) > 0$ for $x \neq x^*$
- (D3) nonincreasing along orbits: $t_0 < t_1 \Rightarrow L(\varphi_{t_0}(x_0)) \geq L(\varphi_{t_1}(x_0))$ (*strict*: never $=$)

differentiable case: $\frac{d}{dt}L(\varphi_t(x)) = \nabla L \cdot f \leq 0$ (Lie derivative)

Lyapunov criterion & LaSalle
$\exists$ Lyapunov function $\Rightarrow$ **stable** (sublevel sets S_ε are positively invariant)
L not constant on any nontrivial orbit $\Rightarrow$ **asymptotically stable** (L is constant on $\omega(x)$, so $\omega(x) = \{x^*\}$)

ω-/α-limit sets
$\omega(x) = \{y : \exists t_j \to +\infty,\ \varphi_{t_j}(x) \to y\}$, $\alpha(x)$: $t_j \to -\infty$

First integral constant of motion
$\nabla g \cdot f \equiv 0$. Hamiltonian: $x' = \partial_y H$, $y' = -\partial_x H$ has $H = \frac{1}{2}|y|^2 + V(x)$ as first integral (not a Lyapunov function in general).
harmonic oscillator: $L = x^2 + y^2$ is both

Linear(ized) stability
$x' = Ax$: all $\mathrm{Re}\,\lambda_j < 0 \Rightarrow$ asympt. stable; some $\mathrm{Re}\,\lambda_j > 0 \Rightarrow$ unstable.
maps $x_{k+1} = Ax_k$: multipliers $|\mu_j| < 1 \Rightarrow$ asympt. stable.
$\mathrm{Re}\,\lambda_j = 0$ cases undecided by linearization → center manifold

Worked example: 1D bifurcation analysis exam pattern
$x' = f(x,p) = px + 2x^2 - x^3 = x\,(p + 2x - x^2)$
- **steady states:** $x = 0$ and $x_\pm = 1 \pm \sqrt{1+p}$ (exist for $p \geq -1$)
- **linear stability:** $\partial_x f = p + 4x - 3x^2$; at $x = 0$: $\lambda = p$ — stable $p < 0$, unstable $p > 0$
- **fold** at $(x,p) = (1,-1)$: $f = 0$, $\partial_x f = 0$, $\partial_{xx} f = -2 \neq 0$, $\partial_p f = 1 \neq 0$ ✓
- **transcritical** at $(0,0)$: x_- crosses 0, stability is exchanged
- sketch: branches $x = 0$ and the parabola $p = x^2 - 2x$ in the (p,x)-plane, solid = stable, dashed = unstable

node $b_j = 0,\ a_1a_2 > 0$ **saddle** $b_j = 0,\ a_1a_2 < 0$ **focus** $b_1 \neq 0,\ a_1 \neq 0$ **center** $a_j = 0,\ b_1 \neq 0$
$\lambda_{1,2} = a_{1,2} + ib_{1,2}$ eigenvalues of $Df(x^*)$; qualifiers: stable/unstable by sign a_j.

The equivalence ladder §4

1 h homeomorphism mapping orbits to orbits
phase portraits agree as sets of curves

2 preserves direction of time

3 preserves time parametrization $h \circ \varphi_t = \psi_t \circ h \quad \forall t$
maps: $f = h^{-1} \circ g \circ h$

4 $h, h^{-1} \in C^k$ (diffeomorphism)
flows: $f(x) = (Dh(x))^{-1} g(h(x))$

- **topological equivalence**: preserves: equilibria, periodic orbits, stability
- **topological conjugacy**: + periods, entropy; Hartman–Grobman lands here (locally)
- **C^k-equivalence**: rows 1–2 + 4
- **C^k-conjugacy**: + eigenvalues / multipliers of linearization

Conjugacy in action
tent map $\cong$ logistic map $4y(1-y)$ via $h(x) = \frac{1-\cos\pi x}{2}$: $f \circ h = h \circ g$.
to transport a system: $y = h(x) \Rightarrow y' = Dh(h^{-1}(y))\,f(h^{-1}(y))$

Hyperbolicity & invariant manifolds §3, §10

flows: x^* hyperbolic $\iff$ $Df(x^*)$ has no eigenvalue with $\mathrm{Re}\,\lambda = 0$

maps: x^* hyperbolic $\iff$ all multipliers $|\mu| \neq 1$ of $Dg(x^*)$

Invariant splitting
$E^s = \{y : e^{tA}y \to 0,\ t \to +\infty\}$, E^u: $t \to -\infty$; $\mathbb{R}^d = E^s \oplus E^u$ with projections P^s, P^u.

Stable/unstable manifold theorem flows & diffeos, $f \in C^k$
$$W^s_{\mathrm{loc}}(x^*) = \{y \in U : \varphi_t(y) \in U\ \forall t \geq 0,\ \varphi_t(y) \to x^*\}$$
$W^{s/u}_{\mathrm{loc}}$ are C^k manifolds, **tangent to** $E^{s/u}$ at x^*; globalize by $W^{s/u} = \bigcup_t \varphi_t(W^{s/u}_{\mathrm{loc}})$.
constructions: Lyapunov–Perron (variation of constants + contraction) or Hadamard graph transform; maps: $W^s_{\mathrm{loc}} = \{y : g^k(y) \to x^*\}$
ex. Duffing $\ddot{x} = x - px^3$: saddle at 0, $\dim E^s = \dim E^u = 1$; Arnold cat map on $\mathbb{T}^2$: every point hyperbolic

Hartman–Grobman
x^* hyperbolic $\Rightarrow \varphi_t$ is **locally topologically conjugate** to the linearization e^{tA}, $A = Df(x^*)$.
Maps: $h \circ A = g \circ h$, $h = \mathrm{Id} + H$ unique via Banach fixed point.
$\Rightarrow$ near hyperbolic points, the linear picture is the whole local picture (up to homeomorphism — see equivalence ladder)

Non-hyperbolic? — center manifolds

Spectral split
d_c eigenvalues $\mathrm{Re}\,\lambda = 0$, d_s with $\mathrm{Re}\,\lambda < 0$, d_u with $\mathrm{Re}\,\lambda > 0$; E^c = span of critical eigenvectors.

Center manifold theorem $f \in C^k$
$\exists$ local C^k, d_c-dim invariant manifold $W^c_{\mathrm{loc}}(0)$ tangent to E^c; solutions staying near 0 for $t \to \pm\infty$ lie on it.
caveats: not unique, possibly not analytic/not C^∞ — e.g. $x' = x^2$, $y' = -y$ has the family $y = p\,e^{1/x}$ $(x < 0)$

Reduction theorem
With blocks $x' = Bx + F(x,y)$, $y' = Cy + G(x,y)$ $(x \in \mathbb{R}^{d_c})$: locally topologically equivalent to
$$x' = Bx + F(x, H(x)), \qquad y' = Cy$$
where $W^c_{\mathrm{loc}} = \mathrm{graph}\,H$, $H = \mathcal{O}(x^2)$.

Recipe: computing W^c exam workhorse
- shift equilibrium to 0, split into (B,C) blocks (diagonalize)
- ansatz $y = h(x) = a_2x^2 + a_3x^3 + \ldots$
- invariance equation $\mathcal{N}(h) = Dh(x)\,[Bx + F(x,h)] - Ch - G(x,h) = 0$
- match coefficients; approximation thm: $\mathcal{N}(h) = \mathcal{O}(|x|^p) \Rightarrow H = h + \mathcal{O}(|x|^p)$

ex. $x' = xy$, $y' = -y + px^2$: $a_2 = p$, $W^c: y = px^2 + \mathcal{O}(x^4)$, reduced flow $x' = px^3 + \mathcal{O}(x^5)$.

Flows ↔ maps §5

continuous time φ_t	discrete time g^k
equilibrium $f(x^*) = 0$	fixed point $g(x^*) = x^*$
asympt. stable $\iff \mathrm{Re}\,\lambda_j < 0$	$\iff \lvert\mu_j\rvert < 1$
periodic orbit Γ, period T	fixed point of Poincaré map P
time-T map φ_T	any diffeo, via suspension flow

Poincaré map at $y \in \Gamma$
section $\Sigma = \{x : (x-y)\cdot f(y) = 0\}$ transverse to the flow. Implicit function theorem $\Rightarrow$ return time $\tau \in C^1$, $\tau(y) = T$, and
$$P(x) = \varphi_{\tau(x)}(x) \in C^1(\Sigma \cap B(y;\delta))$$

Multipliers & monodromy
multipliers $\mu_1, \ldots, \mu_{d-1}$ = eigenvalues of $DP(0)$ — independent of the point on Γ, the section, and coordinates (local Poincaré maps are conjugate).
variational equation $u' = Df(\Gamma(t))\,u$; monodromy matrix $M(T)$ has eigenvalues $\{1, \mu_1, \ldots, \mu_{d-1}\}$.
all $|\mu_j| < 1 \Rightarrow \Gamma$ asymptotically stable.

Suspension & obstruction
$g(x) = -\frac{1}{2}x$ is *not* a time-T map of any flow on $\mathbb{R}$ (orientation). But every diffeo g of $\mathbb{R}^d$ is the return map of a flow on the $(d{+}1)$-dim mapping torus $M = (\mathbb{R}^d \times [0,T])/\!\sim$, $(g(x),0) \sim (x,T)$.

Hyperbolic sets §7

Hyperbolic set $g \in C^1$ diffeo, Λ compact invariant
$\exists C > 0$, $\lambda \in (0,1)$ and subspaces $E^s(x), E^u(x) \subset T_xM$:
- (D1) invariance: $Dg_x E^{s/u}(x) = E^{s/u}(g(x))$
- (D2) $|Dg^j_x v| \leq C\lambda^j |v|$ on E^s, $j \geq 0$
- (D3) $|Dg^{-j}_x v| \leq C\lambda^j |v|$ on E^u
- (D4) splitting: $T_xM = E^s(x) \oplus E^u(x)$

Anosov diffeomorphism: $\Lambda = M$. ex. cat map $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ on $\mathbb{T}^2$.

Shadowing
δ-pseudo-orbit: $d(g(x_k), x_{k+1}) \leq \delta$. Near a hyperbolic Λ: $\forall\varepsilon\ \exists\delta$ such that every δ-pseudo-orbit is ε-**shadowed** by a true orbit.
numerics with rounding errors still trace real orbits

Persistence
C^1-small perturbations f of g: nearby compact invariant sets remain hyperbolic, and $\exists$ hyperbolic K with $\psi \circ f|_K = g|_\Lambda \circ \psi$, ψ homeo near Id.
Cor: Anosov diffeos are open in C^1 and structurally stable.
the horseshoe's invariant set is hyperbolic $\Rightarrow$ chaos persists → see the road to chaos

Structural stability §6

Definition
$f \in C^1(M)$ is **structurally stable** if $\exists\varepsilon > 0$: every g with $\|f - g\|_{C^1} < \varepsilon$ generates a flow topologically equivalent to φ^f_t.
$\|f\|_{C^k} = \sum_{|\alpha|\leq k} \sup_K |D^\alpha f|$ on compact K — a Banach space

Nonwandering set
$x \in \mathrm{NW}(\varphi)$: every neighborhood $U \ni x$, $\forall t_0\ \exists t > t_0$: $\varphi_t(U) \cap U \neq \emptyset$.

Peixoto compact 2-manifold
$f \in C^1$ structurally stable $\iff$
- (T1) finitely many equilibria & periodic orbits, all hyperbolic
- (T2) no saddle–saddle connections: $W^s(x) \cap W^u(y) = \emptyset$
- (T3) $\mathrm{NW}(\varphi) = \{\text{equilibria}\} \cup \{\text{periodic orbits}\}$

M orientable $\Rightarrow$ structurally stable fields are **open and dense** in $C^1(M)$. Fails for $d \geq 3$.
ex. torus flow $x' = \omega_1$, $y' = \omega_2$: structurally *unstable* ($\omega_1/\omega_2 \in \mathbb{Q}$: all orbits periodic; irrational: all orbits dense)

Morse–Smale system
(D1) finitely many equilibria + periodic orbits, all hyperbolic; (D2) all $W^s \cap W^u$ transversal; (D3) NW = equilibria $\cup$ periodic orbits.
Thm: Morse–Smale flows on compact M are structurally stable (any dim) — but *not generic* for $d \geq 3$.

Generic property
holds on a **residual** set = countable intersection of open dense sets.
ex. hyperbolicity is generic in $\mathbb{R}^{d\times d}$

What is chaos? §15

1 Sensitive dependence $\exists r > 0\ \forall\delta\ \exists y_0, T$: $|x_0 - y_0| < \delta$, $|\varphi_T(x_0) - \varphi_T(y_0)| \geq r$

2 Topological transitivity some forward orbit is dense in X

3 Chaotic := 1 ∧ 2 on an invariant set S

4 Positive Lyapunov exponent $l_j(x_0) = \lim_k \frac{1}{k}\ln\lambda^{(k)}_j$
$\lambda^{(k)}_j$ singular values of $Dg^k_{x_0}$; Liouville: $\sum_j l_j = \ln|\det Dg|$ (if constant)

- **doubling**: $2x$ mod 1; $l = \ln 2$
- **tent ≅ logistic**: $p{=}2$ / $p{=}4$: conjugate! $l = \ln 2$
- **shift σ**: on Σ_2; universal model; row 4 n/a
- **horseshoe**: $l_1 = \ln\mu$, $l_2 = \ln\lambda$
- **rotation**: irrational: dense orbits, *no* sensitivity

The road to chaos §14, §15

Periodically perturbed Hamiltonian system
$x' = f(x) + \varepsilon g(x,t)$, $f = (\partial_{x_2}H, -\partial_{x_1}H)^\top$, g T-periodic; unperturbed homoclinic orbit $\gamma(t)$ to a saddle p. Autonomize on $\mathbb{R}^2 \times S^1$, take the stroboscopic Poincaré map $P^{t_0}_\varepsilon$ — the saddle persists: $p^{t_0}_\varepsilon = p + \mathcal{O}(\varepsilon)$.

↓ *leading-order splitting of W^s, W^u*

Melnikov function
$$M(t_0) = \int_{-\infty}^{\infty} f(\gamma(t-t_0)) \wedge g(\gamma(t-t_0), t)\,dt$$
signed distance: $d(t_0) = \varepsilon\,\dfrac{M(t_0)}{|f(\gamma(0))|} + \mathcal{O}(\varepsilon^2)$, $a \wedge b = a_1b_2 - a_2b_1$.
simple zero of M $\Rightarrow W^s(p_\varepsilon) \pitchfork W^u(p_\varepsilon)$; no zeros $\Rightarrow$ no intersection.
Duffing $x'' = x - x^3 + \varepsilon(\alpha\cos t - x')$, $\gamma_\pm = \pm(\sqrt{2}\,\mathrm{sech}\,t,\ -\sqrt{2}\,\mathrm{sech}\,t\tanh t)$: $M(t_0) = -\frac{4}{3} + \sqrt{2}\,\alpha\pi\,\mathrm{sech}(\frac{\pi}{2})\sin t_0$ — simple zeros for α large enough

↓ *transverse homoclinic point $q \in W^s(0) \pitchfork W^u(0)$*

Smale–Birkhoff homoclinic theorem
diffeo with hyperbolic saddle + transverse homoclinic point $\Rightarrow$ some iterate g^κ has an invariant set X conjugate to a **horseshoe**.
λ-lemma: a disk D transverse to W^s has iterates $g^n(D)$ accumulating C^1 on W^u — this folds a rectangle near q into a horseshoe. (Transverse homoclinic points occur only in maps / Poincaré maps, never in planar flows.)

↓ *stretch $\mu > 2$, contract $\lambda < \frac{1}{2}$, fold*

Smale horseshoe ≅ full shift
$X = \{x : g^k(x) \in Y = [0,1]^2\ \forall k \in \mathbb{Z}\}$; itinerary map $\varphi(x)_j = a_j$ with $g^j(x) \in H_{a_j}$ is a homeomorphism conjugating $g|_X$ to the shift σ on $\Sigma_2 = \{0,1\}^{\mathbb{Z}}$.
generalized horseshoe: N strips + cone conditions $\Rightarrow$ shift on N symbols

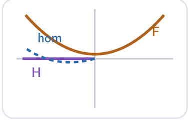
↓ *count orbits symbolically*

Consequences
- countably many (saddle) periodic orbits, of every period
- uncountably many aperiodic orbits; a dense orbit
- X is a hyperbolic set $\Rightarrow$ the whole picture **persists** under perturbation
- $g|_X$ is chaotic; Lyapunov exponents $\ln\mu > 0 > \ln\lambda$

alternative entrance: Shilnikov saddle-focus loop (zoo) also forces horseshoes

The bifurcation zoo §8, §12, §13

Bifurcation loss of equivalence under parameter variation
Families $x' = f(x,p)$, $y' = g(y,q)$ are **fiber topologically equivalent** if a homeomorphism $\xi(p) = q$ matches parameters and each $f(\cdot,p) \sim g(\cdot,\xi(p))$ topologically. A **bifurcation diagram** stratifies parameter space by equivalence class; an **unfolding** contains the bifurcation persistently; the minimal number of parameters is the **codimension**.

eigenvalues bifurcation	genericity conditions	topological normal form	what happens
fold (saddle-node) flows · codim 1	$f(0,0) = 0$, $\partial_x f(0,0) = 0$; $\partial_{xx} f(0,0) \neq 0$, $\partial_p f(0,0) \neq 0$	$y' = q \pm y^2$	two equilibria collide and annihilate; $x' = p - x^2$: $\pm\sqrt{p}$ for $p > 0$, none for $p < 0$
transcritical flows · constrained	$\lambda = 0$ with fixed line of equilibria $x = 0$	$y' = y(q - y)$	two equilibria cross and **exchange stability**
Hopf flows, $d \geq 2$ · codim 1	$\lambda_{1,2}(p) = \mu(p) \pm i\omega(p)$, $\mu(0) = 0$, $\omega(0) > 0$; first Lyapunov coeff. $l_1 \neq 0$	$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} q & -1 \\ 1 & q \end{pmatrix} y \pm (y_1^2 + y_2^2)\,y$	**limit cycle** born; $l_1 < 0$ supercritical (stable cycle), $l_1 > 0$ subcritical
fold of maps maps · codim 1	$\mu = 1$; $\partial_{xx} g \neq 0$, $\partial_p g \neq 0$	$y \mapsto q + y \pm y^2$	fixed points collide and annihilate
flip (period doubling) maps · codim 1	$\mu = -1$; non-degeneracy + transversality	$y \mapsto -(1+q)\,y \pm y^3$	fixed point loses stability, stable **period-2 orbit** emerges; cascades → Feigenbaum
Neimark–Sacker maps · codim 1	complex pair of multipliers crosses the unit circle	(normal form on W^c, rotation + cubic)	closed **invariant curve** bifurcates; Hénon map $(x_1,x_2) \mapsto (x_2,\ p_1 - p_2x_1 - x_2^2)$: NS, flip & fold regions
Bogdanov–Takens flows · codim 2	double zero eigenvalue, nilpotent $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$	$y_1' = y_2$, $y_2' = q_1 + q_2y_1 + y_1^2 \pm y_1y_2$	 homoclinic loop; fold $4q_1 = q_2^2$, Hopf $q_1 = 0$, $q_2 < 0$, homoclinic curve between — rescaling → Hamiltonian $\frac{1}{2}Y_2^2 - \frac{1}{2}Y_1^2 + \frac{1}{3}Y_1^3$ with
homoclinic loop Andronov–Leontovich $\mathbb{R}^2$ · global, codim 1	homoclinic orbit to saddle, $\lambda_1 < 0 < \lambda_2$; saddle quantity $\sigma = \lambda_1 + \lambda_2 \neq 0$; split function $\xi'(0) \neq 0$	$\xi(p)$ = signed distance of W^u, W^s on a section Σ near the saddle	unique **limit cycle** bifurcates from the loop; $\sigma < 0$: stable, $\sigma > 0$: unstable
Shilnikov $\mathbb{R}^3$ · global	homoclinic orbit to saddle-focus: $\lambda_1 > 0$, $\lambda_{2,3} = \alpha \pm i\omega$, $\alpha < 0$; $\sigma = \lambda_1 + \alpha > 0$, $\lambda_2 \neq \lambda_3$	spiral return map intersects itself transversally	**infinitely many** saddle periodic orbits accumulate on the loop $\Rightarrow$ chaos; ex. FitzHugh–Nagumo traveling waves, $0 < \varepsilon \ll 1$

Homoclinic orbit: $\gamma(t) \to x^*$ for $t \to \pm\infty$; heteroclinic: different limits. Structurally unstable in $\mathbb{R}^2$ (Peixoto) but persistent phenomena occur for $d \geq 3$.

Normal forms: the reduction pipeline §9, §10, §11

non-hyperbolic equilibrium $d_c > 0$ critical eigenvalues block B; $A = Df(0)$ → **center-manifold reduction** restrict to W^c_{loc}: dynamics in $\mathbb{R}^{d_c}$ decide everything → **kill non-resonant terms** $x = y + h^{(k)}(y)$, solve $Lh^{(k)} = f^{(k)}$ order by order → **resonant normal form** $y' = Ay + r^{(2)} + \cdots + r^{(k)} + \mathcal{O}(|y|^{k+1})$ → **truncate + unfold** add parameters q → topological normal forms of the zoo

Homological equation
$L: H_k \to H_k$ on homogeneous vector polynomials, $Lh = Dh \cdot Ay - A\,h(y) = [h, Ay]$.
Solvable part of $f^{(k)}$ is removed; **resonant terms** span a complement $R_k = \ker L^*$ of range L (if no generalized 0-eigenvectors: $H_k = \ker L \oplus \mathrm{range}\,L$).
in practice: write L as matrix $M^{(k)}$ in monomial basis, e.g. $k = 2$: $\{x^2, xy, y^2\} \otimes \{\partial_x, \partial_y\} \cong \mathbb{R}^6$; resonant terms from $\ker(M^{(k)})^\top$

Hopf normal form eigenvalues $\pm i\omega$
$$y' = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} y + (y_1^2 + y_2^2)\begin{pmatrix} p_1 & -p_2 \\ p_2 & p_1 \end{pmatrix} y + \mathcal{O}(|y|^4)$$
cubic terms are resonant — they cannot be removed; $p_1 = l_1$ decides super/subcritical

Jets & determinacy §11
k-jet J^kF = Taylor polynomial to order k. BT 2-jet $y_1' = y_2$, $y_2' = y_1^2 \pm y_1y_2$ is **topologically determinate**: blow-up $\Phi(\theta,r) = (r\cos\theta, r\sin\theta)$, desingularize $\bar{F} = r^{-l}(D\Phi^{-1} \circ F \circ \Phi)$ — the origin blows up to a circle of hyperbolic equilibria, so Hartman–Grobman finishes the classification.

One-dimensional maps §16

$1 \lhd 2 \lhd 4 \lhd 8 \lhd \cdots \lhd 2^n{\cdot}7 \lhd 2^n{\cdot}5 \lhd 2^n{\cdot}3 \lhd \cdots \lhd 9 \lhd 7 \lhd 5 \lhd 3$

Sharkovskii $g \in C^0([a,b],\mathbb{R})$
periodic point of minimal period $n \Rightarrow$ periodic orbits of every period $k \lhd n$.
Li–Yorke: period three $\Rightarrow$ all periods ("period three implies chaos").
tent map $p = 2$: $\frac{2}{7} \mapsto \frac{4}{7} \mapsto \frac{6}{7} \mapsto \frac{2}{7}$ is a 3-orbit

Schwarzian derivative
$$Sg = \frac{g'''}{g'} - \frac{3}{2}\left(\frac{g''}{g'}\right)^2$$
minimum principle: $Sg < 0 \Rightarrow |g'|$ has no interior local minimum.
unimodal: $g(0) = g(1) = 0$, unique max; S-unimodal: also $Sg < 0$

Singer
$g \in C^3(I,I)$, $Sg < 0$, n critical points $\Rightarrow$ at most $n+2$ attracting periodic orbits.
logistic $px(1-x)$: $Sg = \frac{-6}{(1-2x)^2} < 0$, one critical point $\Rightarrow\ \leq 3$

Renormalization & universality §17

Idea
act on maps by *rescaling* phase space, time (iterate), and parameters; self-similar dynamics = fixed point of the renormalization operator.
fold: $g^2(y;q) \approx 2q + y - 2y^2$ rescales back to a fold $(\alpha = 2, \beta = 4) \Rightarrow$ passage time $N(q) \sim q^{-1/2}$ (critical slowing down)

Feigenbaum universality S-unimodal maps
flip cascade $p_0 < p_1 < \cdots \to p_\infty$ (logistic: $p_0 = 3$, 2-cycle multiplier $4 + 2p - p^2$),
$$\delta_F = \lim_n \tfrac{p_n - p_{n-1}}{p_{n+1} - p_n} \approx 4.6692, \qquad d_n = \kappa_2\,\alpha_F^{-n}, \quad \alpha_F \approx -2.5029$$
d_n: distance of the superstable 2^n-cycle to the critical point $x = \frac{1}{2}$ (superstable: all multipliers $= 0$, i.e. $\frac{1}{2}$ in the cycle)
fixed-point equation $g(x) = \alpha\,g^2(x/\alpha)$, $g(0) = 1$, $g'(0) = 0$; δ_F = the unique unstable eigenvalue of the linearized operator — the same constants for *every* such family.

Attractors §18

0 Trapping region U compact, $\varphi_t(U) \subset \mathrm{int}(U)\ \forall t > 0$

1 Attracting set $A = \bigcap_{t\geq 0} \varphi_t(U)$
closed, invariant; $\omega(x_0) \subset A$ on U; hyperbolic $x^* \in A \Rightarrow W^u(x^*) \subset A$

2 Indecomposable no smaller attracting set inside

3 Chaotic on A sensitive + transitive

4 Hyperbolic set uniform splitting (D1)–(D4)

- **attracting set**: $r' = 1 - r$ on circle: $\{r{=}1\}$ attracts, but most orbits go to one point
- **attractor**: stable equilibria, stable limit cycles
- **chaotic / strange attractor**: **strange** = chaotic ∧ *not* hyperbolic: Lorenz
- **hyperbolic attractor**: Smale–Williams solenoid: locally Cantor × interval

Lorenz system $\sigma = 10$, $\rho = 28$, $\beta = 8/3$
$x_1' = \sigma(x_2 - x_1)$, $x_2' = \rho x_1 - x_2 - x_1x_3$, $x_3' = x_1x_2 - \beta x_3$; the ellipsoid $\sigma x_1^2 + x_2^2 + \beta(x_3 - \frac{\rho+\sigma}{2})^2 \leq c$ is trapping. $\nabla\cdot f = -(\sigma + 1 + \beta) < 0 \Rightarrow$ the attractor has **volume zero**.

Smale–Williams solenoid hyperbolic attractor
$g(\theta,x,y) = \left(2\theta,\ \lambda x + \frac{1}{2}\cos 2\pi\theta,\ \lambda y + \frac{1}{2}\sin 2\pi\theta\right)$ on $S^1 \times D^2$, $\lambda \in (0,\frac{1}{2})$; $A = \bigcap_k g^k(S^1 \times D^2)$.

Milnor attractor measure-theoretic
closed invariant A with $\mu(B(A)) > 0$ and no proper closed $\tilde{A} \subset A$ with $\mu(B(A) \setminus B(\tilde{A})) = 0$.
ex. $x' = -x^2$: $x^* = 0$ attracts $[0,\infty)$ although unstable

Fractal dimension
box-counting: $\dim_B(S) = \lim_{r\to 0} \frac{\ln N(r,S)}{\ln(1/r)}$; middle-third Cantor set: $\frac{\ln 2}{\ln 3} \approx 0.63$. Strange attractors typically have non-integer dimension.
Milnor attractor (positive-measure basin) — see stability ladder, row 2

Recurrence & Conley's theorem §19

0 Fixed: $g(x) = x$

1 Periodic: $g^n(x) = x$

2 Nonwandering $\forall U \ni x$, some $\varphi_t(U) \cap U \neq \emptyset$, t arbitrarily large

3 Chain recurrent $\forall\delta > 0$ a δ-pseudo-orbit loops $x \to x$

- $\mathrm{Fix}(g)$
- $\mathrm{Per}(g)$
- $\mathrm{NW}(g)$
- $\mathrm{CR}(g)$: chain classes = equivalence classes of two-way δ-chains

Conley homeo of compact metric X
$\exists$ **complete Lyapunov function** $L \in C(X,\mathbb{R})$:
- (T1) $L(g(x)) < L(x)$ off $\mathrm{CR}(g)$
- (T2) on $\mathrm{CR}(g)$: $L(x) = L(y) \iff x, y$ chain equivalent
- (T3) $L(\mathrm{CR}(g))$ compact, nowhere dense (Cantor-like)

moral: every system = gradient-like part + chain-recurrent part.

Construction via attractor–repellor pairs
dual repellor $A^* = \bigcap_{k\geq 0} g^{-k}(X \setminus \overline{U})$ of an attractor A with isolating U; there is $L_i \in C(X,[0,1])$ with $L_i|_{A_i} = 0$, $L_i|_{A_i^*} = 1$, strictly decreasing elsewhere. Then
$$\mathrm{CR}(g) = \bigcap_i (A_i \cup A_i^*), \qquad L = \sum_i \tfrac{2}{3^i} L_i$$
$L(\mathrm{CR})$ lands in the middle-third Cantor set — hence nowhere dense

Symbolic dynamics: the universal model §1, §13, §22

Shift spaces
Σ_k = bi-infinite sequences over $\{0,\ldots,k-1\}$; shift $(\sigma\omega)_j = \omega_{j+1}$. Periodic orbits = periodic sequences; a sequence containing every finite word has a dense orbit.

One model, many systems
- horseshoe $g|_X \cong \sigma$ on Σ_2 (itineraries through H_0, H_1)
- tent map $p{=}2$: semi-conjugate to Σ_2^+ via binary coding $H(s) = \sum_j s_j 2^{-(j+1)}$
- Bernoulli measure p on Σ_k → the probabilistic model; entropy $-\sum p_j \ln p_j$

prove chaos once for σ, transport it along conjugacies

The ergodic hierarchy §20–22

0 Invariant (probability) measure $\mu(g^{-1}(A)) = \mu(A)\quad \forall A \in \mathcal{F}$
$\Rightarrow \int f\,d\mu = \int f \circ g\,d\mu$

1 Ergodic $g^{-1}(A) = A \Rightarrow \mu(A) \in \{0,1\}$
$\iff$ invariant $f \in L^2$ const a.e. $\iff$ time avg = space avg a.e. $\iff$ $\frac{1}{n}\sum_j \mu(g^{-j}A \cap B) \to \mu(A)\mu(B)$

2 Weak mixing Cesàro-averaged correlations vanish: $\frac{1}{n}\sum_j |\mu(g^{-j}A \cap B) - \mu(A)\mu(B)| \to 0$

3 Strong mixing $\mu(g^{-n}A \cap B) \to \mu(A)\,\mu(B)$
correlations decay: the system forgets its initial state

- **rotation $r \in \mathbb{Q}$**: Lebesgue invariant; not ergodic
- **rotation $r \notin \mathbb{Q}$**: ergodic (Fourier) but **not** weak mixing: pure point spectrum $e^{2\pi i n r}$
- **expanding maps**: αx mod 1, $\alpha \in \mathbb{Z}$, $|\alpha| > 1$; dyadic intervals refine uniformly
- **Bernoulli shift**: (Σ_k, σ, p); entropy $-\sum p_j \ln p_j$

strong mixing $\Rightarrow$ weak mixing $\Rightarrow$ ergodic; all implications strict.

Ergodic theorems

Krylov–Bogolyubov
X compact metric, g continuous $\Rightarrow$ an invariant Borel probability measure exists (empirical averages + Riesz representation)

Gauss map invariant measures can be exotic
$g(x) = \frac{1}{x} - \lfloor\frac{1}{x}\rfloor$ on $[0,1)$ preserves $\mu((a,b)) = \frac{1}{\ln 2}\int_a^b \frac{dy}{1+y}$ — the shift on continued-fraction digits.

Poincaré recurrence
g measure-preserving, $\mu(B) > 0 \Rightarrow$ a.e. $x \in B$ returns to B infinitely often.

Birkhoff $f \in L^1$
$$\tfrac{1}{n}\sum_{j=0}^{n-1} f(g^j(x)) \longrightarrow f^*(x) \ \text{a.e.}$$
f^* invariant, $\int f^* d\mu = \int f\,d\mu$; **ergodic case:** $f^* = \int f\,d\mu$ a.e. — time average = space average.
von Neumann: also $S_nf \to f^*$ in L^p; key lemma: maximal ergodic theorem $\int_{\{F_N>0\}} f\,d\mu \geq 0$

Koopman & Perron–Frobenius operators
$L_gf = f \circ g$ on L^p (isometry); adjoint P_g: $\langle Pf, h\rangle = \langle f, Lh\rangle$.
- ergodic $\iff$ 1 is a simple eigenvalue of L_g
- invertible + continuous spectrum (1 the only eigenvalue) $\Rightarrow$ weak mixing
- strong mixing $\iff \langle P^nf, h\rangle \to \langle f,1\rangle\langle 1,h\rangle$

irrational rotation: eigenfunctions $e^{2\pi i n x}$ span L^2 (pure point) $\Rightarrow$ not weak mixing

Entropy §23

Entropy of a partition → of a map
$$H(\mathcal{A}) = -\sum_j \mu(A_j)\ln\mu(A_j)$$
$$h(g,\mathcal{A}) = \lim_n \tfrac{1}{n} H\Big(\bigvee_{j=0}^{n-1} g^{-j}\mathcal{A}\Big), \qquad h(g) = \sup_{\mathcal{A}} h(g,\mathcal{A})$$
$\mathcal{A} \vee \mathcal{B} = \{A_k \cap B_n\}$: joint refinement; $H(\mathcal{A} \vee \mathcal{B}) = H(\mathcal{A}) + H(\mathcal{B} \mid \mathcal{A})$

Kolmogorov–Sinai & invariance
h is a **conjugacy invariant**; for a generating partition $\mathcal{A}$: $h(g) = h(g,\mathcal{A})$.

Conditional entropy & rules
$H(\mathcal{A} \mid \mathcal{B}) = -\sum_{i,j} \mu(B_j)\,\mu(A_i \mid B_j)\ln\mu(A_i \mid B_j)$
- $H(\mathcal{A} \vee \mathcal{B}) = H(\mathcal{A}) + H(\mathcal{B} \mid \mathcal{A})$; $\mathcal{A} \subseteq \mathcal{B} \Rightarrow H(\mathcal{A}) \leq H(\mathcal{B})$
- $h(g,\mathcal{A}) \leq H(\mathcal{A})$; $h(g,\mathcal{A}) \leq h(g,\mathcal{B}) + H(\mathcal{A} \mid \mathcal{B})$
- g-invariance: $H(g^{-1}\mathcal{A} \mid g^{-1}\mathcal{B}) = H(\mathcal{A} \mid \mathcal{B})$

Why entropy is finer than spectrum
Bernoulli shifts: $h = -\sum p_j \ln p_j$; $p = (\frac{1}{2},\frac{1}{2})$: $h = \ln 2$, $p = (\frac{1}{3},\frac{1}{3},\frac{1}{3})$: $h = \ln 3 \Rightarrow$ **not conjugate**, though spectrally indistinguishable.
entropy ≈ average information produced per iteration

How to read this sheet

The lattice panels
Numbered rows are **properties**; each colored column is a **notion** that satisfies exactly the rows it covers. Longer column = stronger notion; a gap in a column means that property is genuinely skipped (e.g. a Milnor attractor need not be Lyapunov stable).

Color code
local theory · robustness & bifurcations · chaos · statistics · examples & bridges

The three big bridges
- **Hartman–Grobman:** hyperbolic local dynamics = linear dynamics (up to conjugacy)
- **Smale–Birkhoff:** one transverse homoclinic point $\Rightarrow$ full symbolic chaos
- **Birkhoff ergodic:** time averages = space averages (ergodic case)